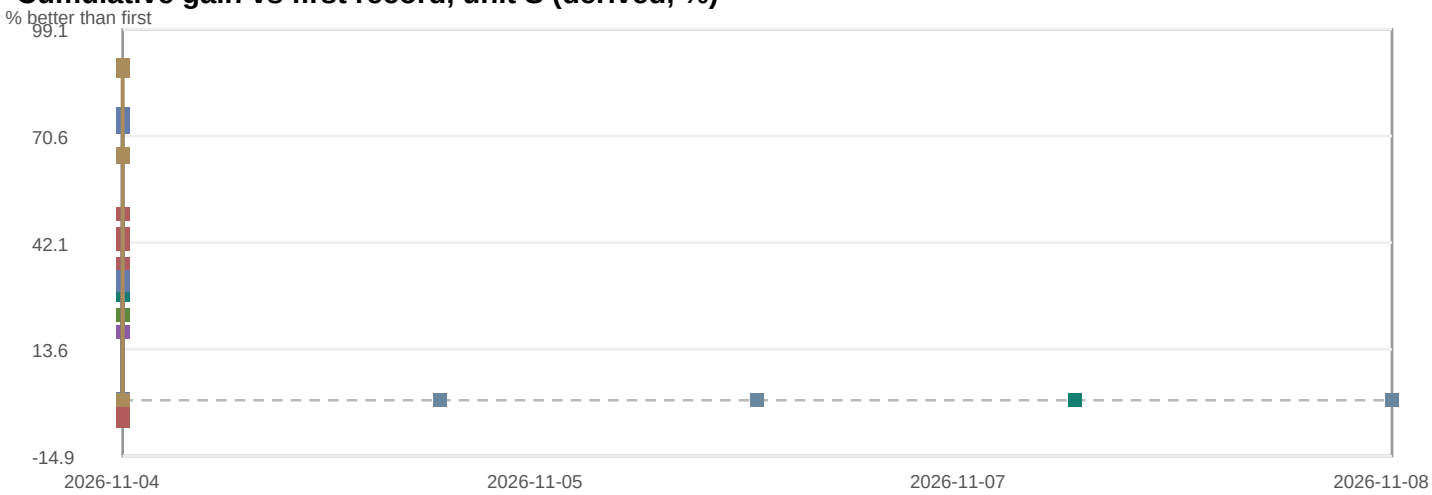
Benchmark progress: crypto

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 5, measured 60, invalid 0, censored 3. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

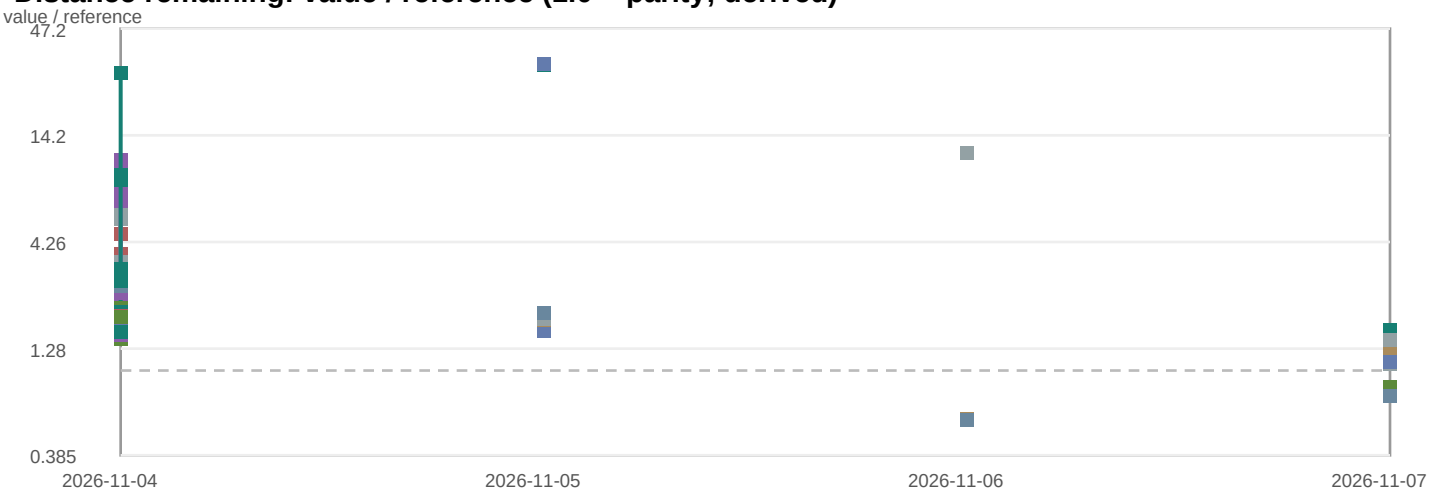
Cumulative gain vs first record, unit S (derived, %)



Series key

- archive:ic-packed-filter-p32-2
- archive:ic-packed-filter-p32-2
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •

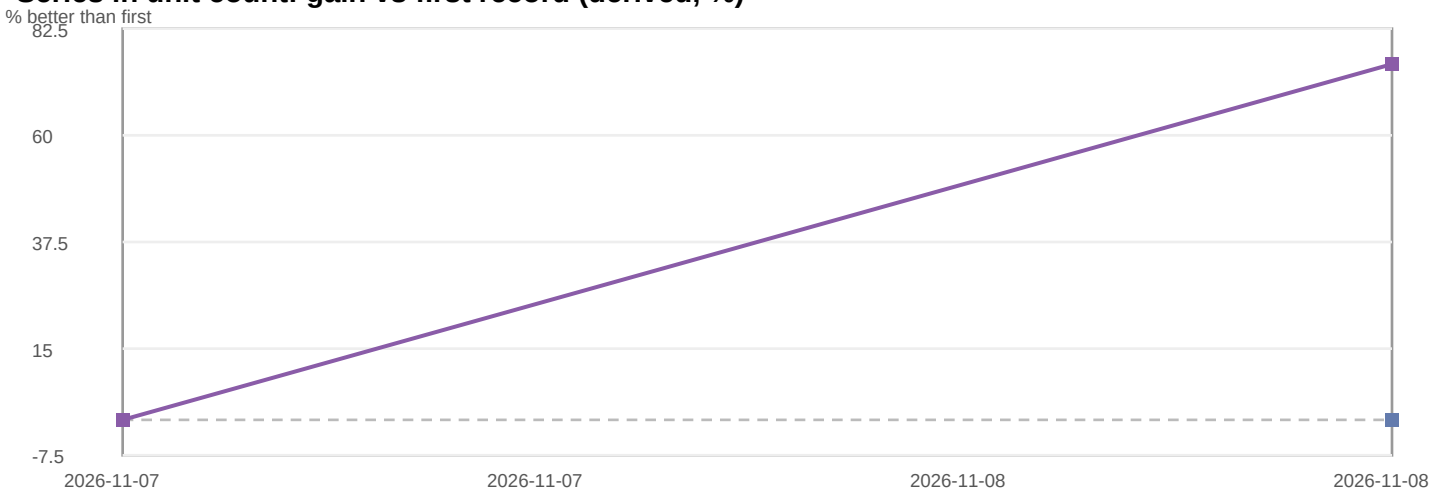
Distance remaining: value / reference (1.0 = parity; derived)



Series key

- archive:ic-packed-filter-p32-2
- archive:ic-packed-filter-p32-2
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe

Series in unit count: gain vs first record (derived, %)



Series key

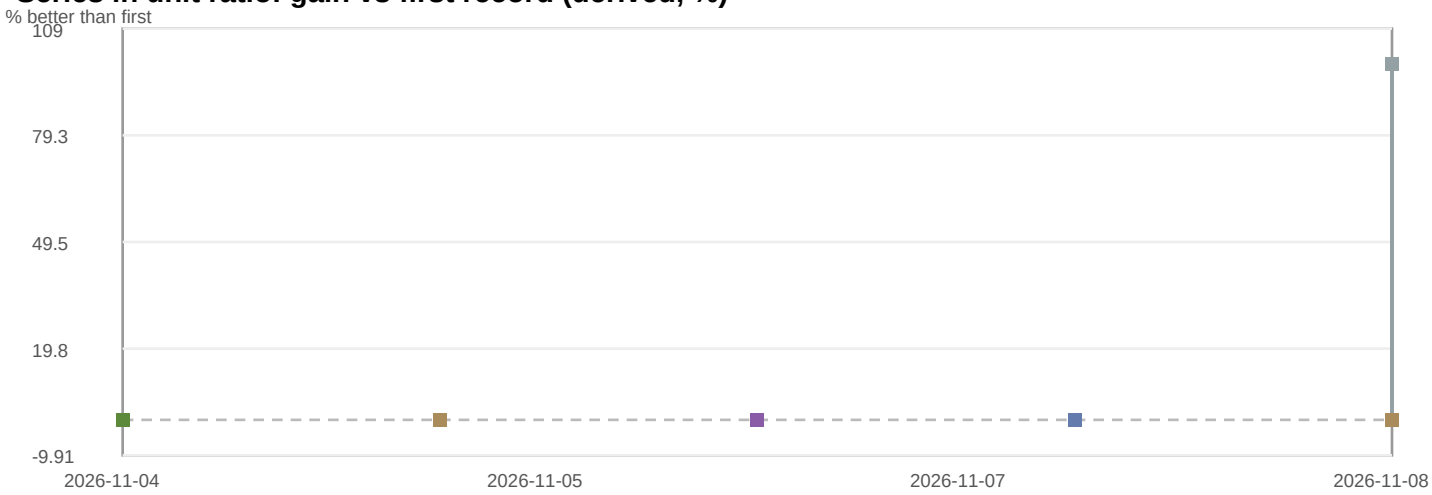
- analysis:ecbench_statistics_v2
- archive:icb-large-prime-n53-s1
- archive:icb-large-prime-n53-s2
- archive:icb-large-prime-n61-s1
- archive:icb-large-prime-n61-s2
- ecdlp:split-prime-p3-cross-tra
- ecdlp:split-prime-p3-cross-tra
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-
- registry-rebinding:ic-aimed-ga
- study:elliptic_code_list_decod

Series in unit percent: gain vs first record (derived, %)

Series key

- ecdlp:split-prime-p3-cross-tra
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-

Series in unit ratio: gain vs first record (derived, %)



Series key

- analysis:ecbench_statistics_v2
- analysis:ecbench_statistics_v2
- archive:ic-packed-filter-p32-2
- ecdlp:split-prime-p3-cross-tra
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-
- kernel:f6_ic_gate • f4_control
- kernel:f6_ic_gate • f6ic_cpu_o
- kernel:f6_ic_gate • f6ic_cpu_o
- kernel:f6_ic_gate • f6ic_word_
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p

Series summary

problem • metric	unit	first	latest	gain vs first	value/ref	pts
analysis:ecbench_statistics_v2 • censored_ratio_	ratio	648.2	648.2	-	-	1
analysis:ecbench_statistics_v2 • heavy_tail_rows	count	71	71	-	-	1
analysis:ecbench_statistics_v2 • ops_interval_wi	ratio	13.1	13.1	-	-	1
archive:ic-packed-filter-p32-20261029 • s_prime_	S	31.6936	31.6936	0%	31.4718	1
archive:ic-packed-filter-p32-20261029 • s_prime_	S	63.4961	63.4961	0%	31.6255	1
archive:ic-packed-filter-p32-20261029 • wall_per	ratio	0.9601	0.9601	0%	-	1
archive:icb-large-prime-n53-s1-20261029 • report	count	240	240	-	-	1
archive:icb-large-prime-n53-s2-20261029 • report	count	240	240	-	-	1
archive:icb-large-prime-n61-s1-20261029 • report	count	72	72	-	-	1
archive:icb-large-prime-n61-s2-20261029 • report	count	60	60	-	-	1
dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1	S	0.5739	0.5739	0%	0.5814	2
dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1	S	12.2426	12.2426	0%	11.6066	2
dlpwai:static-dh-p224-shape • s_w0_cheon_pplus1_	S	0.5819	0.5819	0%	0.5746	2
ecdlp:split-prime-p3-cross-trace-census • aa_rho	ratio	0.986	0.009	99.0872%	-	2
ecdlp:split-prime-p3-cross-trace-census • bonfer	count	0	0	-	-	1
ecdlp:split-prime-p3-cross-trace-census • bonfer	count	0	0	-	-	1
ecdlp:split-prime-p3-cross-trace-census • max_ab	percent	0.7	0.087	87.5714%	-	2
ic:curve25519-family-k20-32 • ratio_w0_k24	ratio	0.9375	0.9375	0%	-	1
ic:curve25519-family-k20-32 • ratio_w0_k32	ratio	0.8802	0.8802	0%	-	1
ic:curve25519-family-k20-32 • s_w0_incumbent_k20	S	2.5746	2.5746	0%	1.7601	1
ic:curve25519-family-k20-32 • s_w0_incumbent_k24	S	2.1638	1.6735	22.6592%	1.429	2
ic:curve25519-family-k20-32 • s_w0_incumbent_k28	S	1.9211	1.5717	18.1849%	1.4943	2
ic:curve25519-family-k20-32 • s_w0_incumbent_k32	S	2.11	1.5152	28.1881%	1.5539	2
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	2.601	2.601	0%	1.5669	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	2.111	2.111	0%	1.7695	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	1.869	1.869	0%	1.7971	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	1.93	1.93	0%	1.9052	1
ic:curve25519-family-k20 • s_w0_incumbent	S	8.3478	4.8598	41.7827%	3.2356	5
ic:curve25519-family-k24-28 • s_w0_incumbent_k24	S	2.5842	2.5842	0%	2.0304	1
ic:curve25519-family-k24-28 • s_w0_incumbent_k28	S	2.3732	2.3732	0%	2.3865	1
ic:curve25519-family-k24-32 • best_train_s_w0_k2	S	2.713	2.713	0%	-	1
ic:curve25519-family-k24-32 • best_train_s_w0_k3	S	2.768	2.768	0%	-	1
ic:curve25519-family-k24-32 • ratio_w0_k32	ratio	0.9106	0.9106	0%	-	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k24	S	2.1055	2.1055	0%	1.942	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k28	S	1.8835	1.8835	0%	1.8154	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k32	S	1.8423	1.8423	0%	1.8232	1
ic:curve25519-family-k24 • s_w0_incumbent	S	6.607	3.3182	49.7782%	2.7163	5
ic:curve25519-family-k28-32 • s_w0_incumbent	S	2.5798	2.5798	0%	2.5891	1
ic:curve25519-family-k28-32 • s_w0_seed_mean_k28	S	1.969	1.969	0%	1.8593	1
ic:curve25519-family-k28-32 • s_w0_seed_mean_k32	S	1.781	1.781	0%	1.8229	1
ic:curve25519-family-k28 • s_w0_incumbent	S	10.9395	2.576	76.452%	2.7572	7
ic:curve25519-family-k32 • s_w0_incumbent	S	24.1377	2.519	89.5641%	2.7453	5
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.2283	1.2283	-0%	-	1
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.2385	1.2385	-0%	-	1
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.1542	1.1542	-0%	-	1
ic:curve25519-k20-single-target • ratio_tw_over_	ratio	1.1656	1.1656	0%	-	1
ic:curve25519-k20-single-target • ratio_twcj_ove	ratio	0.8872	0.8872	0%	-	1
ic:curve25519-k20-single-target • ratio_twcj_ove	ratio	0.9088	0.9088	0%	-	1
ic:curve25519-k20-single-target • ratio_twcj_ove	ratio	0.7335	0.7335	0%	-	1
ic:curve25519-k20-single-target • ratio_twcj_ove	ratio	1.3058	1.3058	0%	-	1
ic:curve25519-k20-single-target • ratio_twcj_ove	ratio	1.3963	1.3963	0%	-	1
ic:curve25519-k20-single-target • ratio_twcj_ove	ratio	1.3602	1.3602	0%	-	1
ic:curve25519-k20-single-target • ratio_twcj_ove	ratio	0.8312	0.8312	0%	-	1
ic:curve25519-k20-single-target • ratio_twcj_ove	ratio	0.7797	0.7797	0%	-	1
ic:curve25519-k20-single-target • s_f1_x07_w01	S	2.3781	2.3781	0%	1.4287	1

ic:curve25519-k20-single-target • s_f1_x07_w0	S	2.3211	2.3211	0%	1.4951	1
ic:curve25519-k20-single-target • s_f1_x07_w1	S	2.8907	2.8907	0%	1.0818	1
ic:curve25519-k20-single-target • s_twcj_w01	S	2.1428	2.1428	0%	1.3059	1
ic:curve25519-k20-single-target • s_twcj_w0	S	2.1383	2.1383	0%	1.3963	1
ic:curve25519-k20-single-target • s_twcj_w1	S	2.183	2.183	0%	0.8311	1
ic:curve25519-k20-single-target • twcj_walk_s	S	1.3895	1.3895	0%	-	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.1475	1.1475	-0%	-	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.1636	1.1636	-0%	-	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.0384	1.0384	-0%	-	1
ic:curve25519-k24-single-target • ratio_tw_over_	ratio	0.9801	0.9801	0%	-	1
ic:curve25519-k24-single-target • ratio_twcj_ove	ratio	0.8609	0.8609	0%	-	1
ic:curve25519-k24-single-target • ratio_twcj_ove	ratio	0.8882	0.8882	0%	-	1
ic:curve25519-k24-single-target • ratio_twcj_ove	ratio	0.675	0.675	0%	-	1
ic:curve25519-k24-single-target • ratio_twcj_ove	ratio	1.2961	1.2961	0%	-	1
ic:curve25519-k24-single-target • ratio_twcj_ove	ratio	1.4114	1.4114	0%	-	1
ic:curve25519-k24-single-target • ratio_twcj_ove	ratio	1.3883	1.3883	0%	-	1
ic:curve25519-k24-single-target • ratio_twcj_ove	ratio	0.7492	0.7492	0%	-	1
ic:curve25519-k24-single-target • ratio_twcj_ove	ratio	0.9063	0.9063	0%	-	1
ic:curve25519-k24-single-target • s_f1_x07_w01	S	1.8945	1.8945	0%	1.4975	1
ic:curve25519-k24-single-target • s_f1_x07_w0	S	1.8341	1.8341	0%	1.5802	1
ic:curve25519-k24-single-target • s_f1_x07_w1	S	2.4383	2.4383	0%	1.1058	1
ic:curve25519-k24-single-target • s_twcj_w01	S	1.6241	1.6241	0%	1.2961	1
ic:curve25519-k24-single-target • s_twcj_w0	S	1.623	1.623	0%	1.4113	1
ic:curve25519-k24-single-target • s_twcj_w1	S	1.6342	1.6342	0%	0.7492	1
ic:curve25519-k24-single-target • twcj_walk_s	S	1.394	1.394	0%	-	1
ic:split-prime-p1-rr-dickson-torus • capped_jobs	count	8	2	75%	-	2
ic:split-prime-p1-rr-dickson-torus • rr_degree_r	count	6	6	0%	-	1
ic:split-prime-p1-rr-dickson-torus • rr_vs_f_wid	ratio	2.29	2.29	0%	-	1
ic:split-prime-p1-rr-dickson-torus • rr_vs_t_wid	ratio	1.33	1.33	0%	-	1
ic:split-prime-p1-rr-dickson-torus • t_degree_re	count	9	9	0%	-	1
ic:split-prime-p2-twist-coset-toy • coverage_law	percent	0.93	0.93	0%	-	1
ic:split-prime-p2-twist-coset-toy • f4_work_rel_	percent	0.39	0.39	0%	-	1
ic:split-prime-p2-twist-coset-toy • f4_work_rel_	percent	0	0	-	-	1
ic:split-prime-p2-twist-coset-toy • gain_min_t4	ratio	2.25	2.25	-0%	-	1
ic:split-prime-p2-twist-coset-toy • purity_nontr	count	0	0	-	-	1
ic:split-prime-p2-twist-coset-toy • unplanted_ov	ratio	0.896	0.896	0%	-	1
ic:split-prime-p2-twist-coset-toy • unplanted_ta	count	0	0	-	-	1
kernel:f6_ic_gate • f4_control_word_xors_off_ove	ratio	1	1	-	-	1
kernel:f6_ic_gate • f6ic_cpu_off_over_on_max	ratio	10.29	10.29	-0%	-	1
kernel:f6_ic_gate • f6ic_cpu_off_over_on_min	ratio	2.67	2.67	-0%	-	1
kernel:f6_ic_gate • f6ic_word_xors_off_over_on_m	ratio	1.84	1.84	-0%	-	1
kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	0.936	0.936	-0%	-	1
kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	2.883	2.883	-0%	-	1
kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	1.589	1.589	-0%	-	1
kernel:inherited_f4_solve_tree • work_par_over_s	ratio	2.865	2.865	0%	-	1
registry-rebinding:ic-aimed-gather • raw_pins_re	count	211	211	-	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	0.718	0.718	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0948	1.0948	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.4723	1.4723	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0653	1.0653	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0008	1.0008	-0%	-	1
rho:prime_medium_tier_bound • prime-medium/bsgs-	S	1.0827	1.0827	0%	-	1
rho:prime_medium_tier_bound • prime-medium/bsgs-	S	0.9842	0.9842	0%	-	1
rho:prime_medium_tier_bound • prime-medium/bsgs-	S	1.9684	1.9684	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	1.0422	1.0422	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	0.9473	0.9473	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	1.8961	1.8961	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	1.0658	1.0658	0%	-	1

rho:prime_medium_tier_bound • prime-medium/rho-n	S	0.969	0.969	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	1.9367	1.9367	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-p	S	1.2703	1.2703	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-p	S	1.2703	1.2703	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-p	S	1.2703	1.2703	0%	-	1
study:elliptic_code_list_decoding_20261107 • d1_	count	0	0	-	-	1

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’ are computed here from those values and are not measurements. Series in different units are never compared.